

Global Terrorism

Patterns & Trends Analysis — Project Documentation

Python · MySQL · Power BI Desktop

163 Countries · 95,993 Fatalities · 2012–2022 · 80.28% Top-10 Concentration

TOOLS	SOURCE	RECORDS	SCOPE
Python · MySQL · Power BI	Global Terrorism Index (GTI)	1,793 · 9 Cols	163 Countries, 11 Years

01 EXECUTIVE SUMMARY

- **Analyzed 1,793 country-year records** from the Global Terrorism Index, spanning 163 countries across 11 years (2012–2022) — a complete panel with zero missing values and zero duplicates, requiring structural rather than corrective cleaning: consolidating a 12-sheet workbook down to the single already-melted 'Combined raw' sheet that contained the full dataset.
- **Answered 8 structured geopolitical questions** covering cumulative impact, concentration, global trend direction, the single worst country-year on record, and two correlation analyses — then reproduced every answer independently in MySQL, including 2 hand-built Pearson correlation formulas.
- **Built a 3-page, 14-visual interactive Power BI dashboard** with 5 DAX measures, an ISO3-coded world map, a dual-axis global trend line, and a dedicated severity/period breakdown page built on a bucketed analytical view — connected directly to MySQL.
- **Surfaced a sharply concentrated and genuinely improving picture:** the top 10 countries account for 80.28% of all 95,993 global fatalities, but the global average GTI score fell from a 2018 peak of 2.630 to 2.168 by 2022 — even as Burkina Faso, Mozambique, and Chad saw the sharpest deteriorations, pointing to a real geographic shift in where terrorism impact is concentrated.

02 PROBLEM STATEMENT

The Global Terrorism Index dataset was distributed as a 12-sheet Excel workbook — 11 individual year sheets, a wide 'Overall Scores' pivot, and one already-complete 'Combined raw' panel containing the same underlying data in long format. Without recognizing this redundancy, a naive analysis risks loading and reconciling the same 1,793 country-year records multiple times across mismatched sheet structures. Separately, the dataset's Rank field only makes sense once it's understood that all zero-score, zero-impact country-years in a given year share the same bottom rank — treating Rank as a clean 1-163 ordinal without that context would misrepresent it. The goal of this project was to correctly consolidate the workbook, understand its ranking convention, and answer 8 concrete questions about where terrorism impact is concentrated and how it has trended over an 11-year window.

03 OBJECTIVES

- **Obj 1:** Identify that the workbook's 11 year-sheets and pivot sheet are redundant with the 'Combined raw' panel, avoiding duplicated loading effort.

- **Obj 2:** Quantify how concentrated global terrorism impact is across countries, not just which countries rank highest.
- **Obj 3:** Characterize the 11-year global trend direction using both incident/fatality totals and the composite GTI score.
- **Obj 4:** Identify which countries improved and deteriorated most between the first and last year of the panel.
- **Obj 5:** Reproduce all findings independently in MySQL, including correlation formulas MySQL doesn't provide natively.
- **Obj 6:** Build a Power BI dashboard using a bucketed analytical view to summarize severity and period patterns without rescanning the full panel.

04 DATASET OVERVIEW

Source & Structure

Attribute	Detail
Dataset Name	Global Terrorism Index (GTI)
File Format	Excel workbook, 12 sheets — only 'Combined raw' used
Records	1,793 rows × 9 columns = 163 countries × 11 years, exactly complete
Missing Values	Zero across all columns
Duplicate Rows	Zero
Zero-Impact Rows	636 rows (35.5%) have Score = 0 — legitimate no-impact years, not missing data
Rank Convention	Max rank varies 93–112 by year — only scored countries get a unique rank; zero-score countries share the bottom rank
Redundant Sheets (not loaded)	11 individual year sheets + 1 wide 'Overall Scores' pivot — same data as Combined raw, reshaped

Key Columns Used in Analysis

Column	Type	Description
iso3c	Text	3-letter ISO country code (used directly for Power BI map)
country	Text	163 unique country names
ranking	Integer	Country's GTI rank within its year (renamed from 'rank', a MySQL reserved word)
score	Decimal	Composite GTI impact score, 0–10
incidents	Integer	Count of terrorist incidents in that country-year
fatalities	Integer	Deaths from terrorism in that country-year (global total: 95,993)
hostages	Integer	Hostages taken in that country-year
year_recorded	Integer	2012–2022 (renamed from 'year', a MySQL reserved word)

05 TOOLS & TECHNOLOGIES USED

Tool	Platform	Purpose in This Project
Python 3	VS Code	Workbook consolidation, panel-completeness checks, correlation analysis
pandas / numpy	Python library	Groupby aggregation, year-over-year score-change computation
MySQL 8	MySQL Workbench	Relational storage with reserved-word-safe schema; SQL validation
Power BI Desktop	Microsoft	3-page dashboard with a native ISO3 filled map and a bucketed severity view
DAX	Power BI native language	5 measures: Total Fatalities, Total Incidents, Avg Score, Total Countries, VIEW Weighted Avg Score

06 DATA CLEANING PROCESS — PYTHON

This dataset needed no row-level cleaning at all — the real work was structural: choosing the right sheet and verifying panel completeness before analysis.

Step	Action	Result
1	Load only the 'Combined raw' sheet, ignore the other 11	1,793 rows × 9 columns, avoiding redundant data loading
2	Confirm zero missing values and zero duplicates	Genuinely clean panel confirmed, not assumed
3	Verify exactly 163 countries appear in every one of the 11 years	Confirmed complete panel, no gaps
4	Confirm Score=0 rows are legitimate zero-impact years, not missing data	636 rows (35.5%) correctly retained, not dropped

07 SQL / MYSQL — VALIDATION LAYER

A single flat table (gti), imported with 2 columns renamed to avoid MySQL reserved words (rank → ranking, year → year_recorded). The import statement required special care: Year is the last column in the source CSV, and an earlier version of this pipeline incorrectly trimmed a Windows line-ending artifact from the wrong column (hostages instead of year_recorded) — caught and corrected before the final import.

Two Correlations, No Native Support

MySQL has no CORR() aggregate. Both correlation questions were implemented manually using the Pearson formula: $r = (n\sum xy - \sum x \sum y) / \sqrt{[(n\sum x^2 - (\sum x)^2)(n\sum y^2 - (\sum y)^2)]}$. Both matched Python exactly: incidents↔fatalities 0.8605, hostages↔fatalities 0.3181.

Analytical VIEW

A vw_gti_summary view buckets country-years into 3 time periods (2012–15, 2016–19, 2020–22) crossed with 5 severity bands (No Impact through Severe). Its grand total of 95,993 total_fatalities was independently cross-checked against the raw table's SUM(fatalities) and matched exactly — confirming the bucketing logic doesn't double-count or drop any rows.

08 DAX MEASURES CREATED

Measure	DAX Formula	Result
Total Fatalities	SUM(gti[fatalities])	95,993
Total Incidents	SUM(gti[incidents])	47,873
Avg Score	AVERAGE(gti[score])	2.44
Total Countries	DISTINCTCOUNT(gti[country])	163
VIEW Weighted Avg Score	SUMX-weighted average from vw_gti_summary	2.44 (cross-check, matches Avg Score exactly)

No percent-formatting risk in this project — none of the 5 measures are ratios, so there's no *100/Percentage-format double-counting trap to guard against here, unlike the other 3 projects in this series.

09 DASHBOARD VISUALS BUILT (14 Visuals, 3 Pages)

Page 1 — Global Overview

#	Visual Type	Fields Used	Insight Delivered
1-4	KPI Cards	Total Fatalities / Total Incidents / Total Countries / Avg Score	95.99K fatalities, 47.87K incidents, 163 countries, 2.44 avg score
5	Dual-Axis Line Chart	X: year_recorded / Y1: Total Fatalities / Y2: Avg Score	Fatalities peak 2015, score peaks 2018, both decline sharply to 2022
6	Filled Map	Location: iso3c / Color: Total Fatalities	ISO3 codes recognized natively — no manual Data Category setup needed
7	Bar Chart (Top 10 filtered)	Y: country / X: Total Fatalities	Iraq leads at 21,286, more than 1.6x the next-highest country
8	Slicer	year_recorded	Filters all Page 1 visuals by year

Page 2 — Country-Level Deep Dive

#	Visual Type	Fields Used	Insight Delivered
1	Scatter + trend line	X: incidents / Y: fatalities (Don't summarize)	Strong upward relationship, corr = 0.86
2	Scatter + trend line	X: hostages / Y: fatalities	Visibly weaker, more scattered, corr = 0.32
3	Table (sorted by change)	country, Score 2012, Score 2022, Score Difference	Recreates the most-improved/most-deteriorated ranking directly
4	Slicer	country	Single-country drill-down across Page 2 visuals

Page 3 — Severity & Period Breakdown (from vw_gti_summary)

#	Visual Type	Fields Used	Insight Delivered
1	KPI Card	VIEW Weighted Avg Score	2.44 — matches Page 1's Avg Score exactly, a built-in cross-check
2	Matrix (conditional color)	Rows: severity_bucket / Cols: period_bucket / Values: total_fatalities	05-Severe accounts for 68,669 of 95,993 total fatalities (71.5%)
3	Bar Chart	X: severity_bucket / Y: total_fatalities	Visual confirmation of the severity concentration

#	Visual Type	Fields Used	Insight Delivered
4	Slicer	period_bucket	Filters the matrix and bar chart by time period

10 KEY INSIGHTS & RESULTS

A. Concentration

- **Terrorism impact is extremely concentrated:** the top 10 countries — led by Iraq (21,286), Afghanistan (12,726), and Nigeria (10,191) — account for 80.28% of all 95,993 global fatalities across the 11-year window.
- **The severity breakdown confirms this at a category level:** country-years classified as '05-Severe' alone account for 68,669 of 95,993 total fatalities (71.5%), despite representing a small fraction of all 1,793 country-year records.

B. Global Trend

- **The picture is genuinely improving, on a lag:** total fatalities peaked in 2015 (10,881) while the average GTI score peaked three years later in 2018 (2.630) — both have declined sharply since, reaching 6,701 fatalities and a 2.168 average score by 2022.
- **The single worst country-year on record is unambiguous:** Iraq in 2016 recorded a perfect score of 10.00, with 1,545 incidents and 4,514 fatalities — the most severe entry anywhere in the 1,793-row panel.

C. What Predicts Fatalities

- **Incident count is a strong predictor of fatalities** ($r = 0.86$) — more attacks reliably means more deaths, the expected relationship, now quantified.
- **Hostage-taking is a comparatively distinct pattern** ($r = 0.32$) — far weaker than incidents, suggesting hostage situations follow different dynamics than lethal attacks generally.

D. Where the Trend Is Moving

- **The most-improved countries are geographically diverse:** Sudan (-5.40), China (-5.01), and Belarus (-4.57) show the largest score declines from 2012 to 2022.
- **The most-deteriorated countries cluster sharply in West/Central Africa:** Burkina Faso (+8.56), Mozambique (+7.04), and Chad (+5.11) lead the deterioration ranking — all five of the top deteriorating countries are West or Central African nations, a real regional shift in where terrorism impact is growing even as the global average improves.

11 CHALLENGES FACED & HOW OVERCOME

Challenge 1 — A 12-Sheet Workbook With Only One Sheet Actually Needed

Problem: The source workbook contained 11 individual year sheets and a wide pivot sheet, all containing the same underlying data as the single 'Combined raw' sheet, reshaped differently.

Solution: Verified the row-count math ($163 \text{ countries} \times 11 \text{ years} = 1,793$, matching Combined raw exactly) before committing to loading only that one sheet, rather than assuming redundancy without checking.

Challenge 2 — A Ranking Field That Isn't a Clean 1–163 Ordinal

Problem: Max rank per year ranges from 93 to 112, not a consistent 163 — initially looked like a data-quality issue.

Solution: Investigated rather than assumed — discovered that only countries with a nonzero score receive a unique rank, while all zero-score countries in a given year share the bottom rank. Documented this convention explicitly rather than treating Rank as directly comparable across years.

Challenge 3 — A Line-Ending Trim Applied to the Wrong Column

Problem: An earlier version of the MySQL import statement applied the Windows `\r` line-ending trim to the `hostages` column instead of `year_recorded`, based on an incorrect assumption about column order.

Solution: Explicitly re-verified the saved CSV's column order programmatically (confirming `year_recorded` is genuinely the last column) before finalizing the `LOAD DATA` statement — a reminder that reserved-word renames and Windows CSV artifacts both need to be checked against the actual file, not assumed from memory.

Challenge 4 — Two Column Names Colliding With MySQL Reserved Words

Problem: Both `rank` and `year` — natural, obvious column names — are reserved words in MySQL and would have caused syntax errors in every query touching them.

Solution: Renamed to `ranking` and `year_recorded` at the schema level from the start, avoiding the need to escape them with backticks in every single query throughout the project.

12 SKILLS LEARNED & TAKEAWAYS

- Recognizing redundant data structures across multiple sheets/exports before loading them, using row-count arithmetic as a verification method.
- Investigating an unusual pattern (inconsistent max rank) rather than either ignoring it or assuming it's an error — it turned out to be a documented, meaningful convention.
- Verifying exact column order programmatically before writing import statements, rather than trusting a remembered or assumed order — directly informed by catching a real bug this way.
- Proactively renaming schema fields that collide with reserved words, rather than discovering and patching the problem query-by-query.
- Distinguishing a lagging indicator from a leading one: fatalities and the composite score peaked in different years (2015 vs. 2018), a nuance a single-metric trend line would have missed.
- Reading a geographic clustering pattern (West/Central Africa dominating the deterioration ranking) as a substantive finding worth naming explicitly, not just listing the raw ranked countries.

13 CONCLUSION

- **Consolidated a 12-sheet workbook down to its one complete panel**, verifying $163 \text{ countries} \times 11 \text{ years} = 1,793$ rows exactly before proceeding, and correctly interpreted a non-obvious ranking convention rather than treating it as a data-quality flaw.
- **Independently validated all 8 project questions in MySQL**, including catching and correcting a line-ending trim applied to the wrong column — a real bug found through the same verification discipline applied throughout.
- **Delivered a 3-page, 14-visual Power BI dashboard**, with a dedicated severity/period page whose analytical view was cross-validated against the raw table's grand total to the fatality, confirming the bucketing logic is exact.
- **Most notably, surfaced a two-speed story the aggregate trend alone would have missed:** the world is genuinely safer on average since 2018, but that improvement is not evenly distributed — West and Central

Africa are moving sharply in the opposite direction, a geographic concentration visible only by naming the outliers rather than stopping at the global average.

KPI Summary

Metric	Value	Metric	Value
Countries Covered	163	Years Covered	2012–2022 (11 years)
Total Fatalities	95,993	Total Incidents	47,873
Top 10 Concentration	80.28%	Severe-Bucket Share	71.5%
Worst Single Country-Year	Iraq 2016 — Score 10.00	Peak Fatality Year	2015 — 10,881
Incidents ↔ Fatalities Corr.	0.8605	Hostages ↔ Fatalities Corr.	0.3181
Most Improved	Sudan (-5.40)	Most Deteriorated	Burkina Faso (+8.56)
Avg Score (2018 peak)	2.630	Avg Score (2022)	2.168
Missing Values	0	Duplicate Rows	0